When On-Chain Flow Signals Work and When They Don't: AMM-Layer Stress Propagation Across Five Stablecoin Episodes

A Provenance-Gated Analysis Using Curve Finance TokenExchange Logs

Anonymous Author(s)

Abstract

Empirical contagion studies routinely report cross-market linkage during crises, but rarely separate genuine *contagion* (a shift in linkage caused by the crisis) from constant *interdependence*. Using Tier-A on-chain Curve `TokenExchange` flow, we apply the Forbes–Rigobon regime test to stablecoin stress and find regime-switching contagion that is sharp and mechanism-specific. For the June 2023 USDT/Curve episode — an *endogenous* pool-imbalance shock — cross-pool flow coupling is statistically indistinguishable from zero in calm conditions ($\hat{\rho} = 0.09$, $n = 95$) and activates during the acute window ($\hat{\rho} = 0.53$, $n = 53$, $p < 10^{-4}$); the calm-to-panic shift is significant (Fisher $z = +2.82$, $p = 0.005$). The coupling peaks at lag 0, consistent with a common-shock mechanism rather than sequential transmission. Critically, none of the four *exogenous* episodes (Terra/LUNA, USDC/SVB, FTX, BUSD) shows this activation — even though all now have genuine Tier-A A/A pool pairs to test — so the null is a mechanism finding, not a data gap. We further show two positive structural results. First, on-chain arbitrage is *stabilizing* in calm markets (flow intensity negatively correlated with price deviation) and acute stress significantly bends this in three of five events, flipping it to *amplifying* where arbitrage capacity is overwhelmed (USDT/Curve, BUSD: Fisher $z > 3.6$) and strengthening it where on-chain liquidity absorbs an off-chain run (FTX: $z = -2.8$). Second, price discovery is on-chain for DeFi-native stress: the Curve pool price leads Binance (+1h $r = 0.28$, $p < 10^{-4}$) and deviates 37× more (12.5% vs 0.3%) for USDT/Curve, while the CEX leads only for the exogenous SVB bank run. Finally, on the machine-learning question, Tier-A on-chain pool state adds genuine stress-detection information over a Tier-B market baseline (up to +0.37 AUROC for Terra/LUNA, where the CEX basis alone is near-useless), yet this within-event signal does not transfer: supervised cross-event prediction collapses to chance under concept shift, while an unsupervised Hidden Markov Model recovers the regime from on-chain state without labels (AUROC up to 0.95). The provenance gate certifies the first and rejects the second. The broad implication: DeFi stress surveillance needs *separate monitoring architectures* for endogenous pool-imbalance events versus exchange-originating or regulatory shocks. A provenance-aware three-gate pipeline underpins every claim, blocking fixture and Tier-B data from reaching paper-level conclusions.

ICAIF '26, New York, NY, USA

ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

CCS Concepts

• **Applied computing** → **Economics**; • **Computing methodologies** → *Machine learning*.

Keywords

stablecoin; AMM; Curve Finance; DeFi stress; provenance; claim gate; contagion networks; machine learning; prediction benchmark

ACM Reference Format:
Anonymous Author(s). 2026. When On-Chain Flow Signals Work and When They Don't: AMM-Layer Stress Propagation Across Five Stablecoin Episodes: A Provenance-Gated Analysis Using Curve Finance TokenExchange Logs. In *Proceedings of 7th ACM International Conference on AI in Finance (ICAIF '26).* ACM, New York, NY, USA, 8 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

Between May 2022 and March 2023, four distinct stablecoin systems experienced significant stress: the algorithmic UST/Terra collapsed; FTX's implosion triggered exchange runs; USDC briefly de-pegged following Silicon Valley Bank's failure; and Curve Finance pools became severely imbalanced in June 2023. Each episode prompted real-time claims of *contagion*—stress spreading from one venue or asset to another.

Most empirical analyses of these events rely on price data alone and treat all data sources as equally reliable. This paper addresses both limitations by introducing a provenance-aware framework that makes data quality an explicit input to every empirical claim, and then holding every result—including a supervised prediction benchmark—to the same evidentiary gate. That discipline is what lets us separate the findings that survive from the one that does not: the same gate certifies three structural results on Tier-A on-chain flow and rejects a predictive lift that fails to replicate out-of-event.

The surveillance gap. A price-based DeFi stress monitor observes only stablecoin basis widening—an aggregate market outcome—and is blind to the flow-level mechanism beneath it. Curve Finance `TokenExchange` logs, by contrast, record every swap with an exact block timestamp and exact token amount, freely available and updated every 12 seconds. This is a direct observation of the trading flow that *produces* the price move, not a derived signal. Our analysis shows that the cross-pool flow coupling during the June 2023 USDT/Curve episode is strong but *contemporaneous* with the price stress rather than leading it (Sections 5 and 8); the contribution is therefore a mechanism-level decomposition of how stress moves through the AMM layer, not an early-warning claim. No existing stablecoin contagion study uses TokenExchange flow as its primary evidence layer.

Contributions. *Methodologically*, we introduce a provenance-aware three-gate pipeline (data-quality tier → feature tier → statistical significance) and a claim-level taxonomy (A/A DEX-flow, A/A settlement, A/B directional, B/B contextual) that caps every cross-venue claim at its lowest tier and blocks fixture data from reaching any conclusion. *Empirically*, on five stress episodes we report four results: (1) regime-switching contagion — USDT/Curve cross-pool coupling rises from $\hat{\rho} = 0.09$ (calm) to 0.53 (panic), Fisher $z = 2.82$, $p = 0.005$ — present only for the endogenous shock, despite all five events carrying genuine Tier-A A/A pairs; (2) arbitrage flipping from stabilizing to amplifying under stress ($z = +3.84$ USDT/Curve, +3.69 BUSD, −2.80 FTX); (3) on-chain price discovery, the Curve pool leading the exchange and deviating 37× more for DeFi-native stress; and (4) a machine-learning result with two sides — Tier-A on-chain pool state adds robust within-event stress-detection information over a Tier-B market baseline (up to +0.37 AUROC), but this signal does not transfer across events under concept shift, so an unsupervised HMM (AUROC 0.92–0.95), not a supervised predictor, is the right tool when labels are scarce.

2 Related Work

Stablecoin stress and AMM microstructure. Stablecoin stability has been studied through runs and reserve adequacy [7, 8, 15, 22], DeFi run dynamics [5], and the economics of safe-asset status [14, 18, 19]; classical theory links funding- and market-liquidity co-movement under stress [4]. AMM microstructure work characterises loss-versus-rebalancing [24] and constant-function pricing [2, 26]; we build on Curve's StableSwap design [10] and treat `TokenExchange` logs as execution-grade evidence in the spirit of tick-data econometrics [1].

On-chain contagion measurement. On-chain network methods have tracked exchange fund flows [23], spillover networks [9], and Tether-driven manipulation [17] using lead-lag and Granger tools [16]. Forbes and Rigobon [13] draw the central distinction — contagion (a crisis-driven *shift* in linkage) versus interdependence — which our regime test makes operational on Tier-A flow. Unlike prior work, we explicitly gate co-movement claims by data provenance rather than conflating execution-grade logs with derived market data.

Methodological rigour in financial AI. A recent ICAIF thread foregrounds trustworthiness over raw performance: interpretability as an adoption requirement [11], scrutiny exposing artefactual model signal [20], and evaluation choices reshaping conclusions [21]. Our provenance gate sits in this lineage — an auditable, claim-blocking discipline measured by how reliably it rejects untrustworthy findings. Random Forests and gradient boosting [3, 6] serve here only as a provenance-ablation harness, not a performance claim.

3 Framework, Data, and Provenance

3.1 Three-Layer Network

We model the stablecoin stress environment as a three-layer directed network (Figure 1a).

Layer 1—CEX markets. Nodes are trading pairs at centralized exchanges (e.g., USDC-Coinbase, USDT-Binance, USDT-Kraken). Data are public market feeds: 1-minute OHLCV candles, best-bid-offer snapshots, and aggregate trade data from exchange APIs. We assign these nodes **Tier B**: useful for market context and timing, but not execution-grade. Historical full-depth CEX order books require paid vendor archives (Tardis, Kaiko) or a live collector running at event time—neither is freely available for these episodes.

Layer 2—AMM pools. Nodes are Curve Finance liquidity pools. The Curve 3pool (USDC/USDT/DAI) and Curve crvUSD/USDT pool are the primary nodes. Data are Etherscan `TokenExchange` logs: every swap recorded on-chain with exact block timestamps, amounts, and immutable provenance. We assign these nodes **Tier A**.

Layer 3—Settlement flows. Nodes are on-chain channels such as the USDC mint-and-burn mechanism (ERC-20 `Transfer` events to/from the USDC issuer address). Data are Etherscan event logs. Tier A.

Edge tier formula. An edge from node i to node j is capped by the weakest link:

$$\text{tier}_{ij} = \min\bigl(\text{tier}_i,\ \text{tier}_j,\ \text{tier}_f\bigr), \tag{1}$$

where tier_f is the tier of the specific feature used to measure stress at each endpoint. A Tier-A pool with a derived proxy feature (e.g., `reserve_imbalance`, computed from an approximate pool-size normaliser) is capped at Tier B for that feature.

3.2 Event Windows and Node Inventory

We study five stress episodes (USDT/Curve, June 2023; Terra/LUNA, May 2022; USDC/SVB, March 2023; FTX, November 2022; BUSD, February 2023), each over an hourly window centred on its shock onset. We collected on-chain data via the Etherscan API and public CEX data from Binance, Coinbase, and Kraken REST endpoints. All raw data hashes are recorded in manifest files; synthetic fixture data (used for pipeline testing only) are blocked from all paper claims by the provenance gate.

4 Claim-Gated Methodology

The pipeline applies three gates in sequence (Figure 1b).

Each method has a defined role: AMM lead-lag is primary; transfer entropy and TVP-VAR are corroborating/robustness; the supervised ML benchmark is a provenance-ablation harness; and the HMM is the unsupervised detector.

Gate 1 (provenance) assigns each edge the effective tier $\min(\text{tier}_i, \text{tier}_j, \text{tier}_f)$ and blocks fixture-derived rows unconditionally. *Gate 2 (statistical)* evaluates the AMM lead-lag cross-correlation of `usdc_net_sold_1h` on a 3600 s grid with maximum lag ±12 h, assessing significance by Bonferroni correction over all 25×5×2 = 250 lag–event–direction tests, with a 1000-sample block bootstrap as an independent check; settlement-layer events use a sparse event-arrival test, and transfer entropy and Granger causality serve as corroborating methods. *Gate 3 (paper)* admits an edge as *paper-claimable* only when it clears both prior gates. The resulting taxonomy ranks evidence as A/A DEX-flow and A/A settlement (paper-claimable), A/B directional (indicative), and B/B contextual.

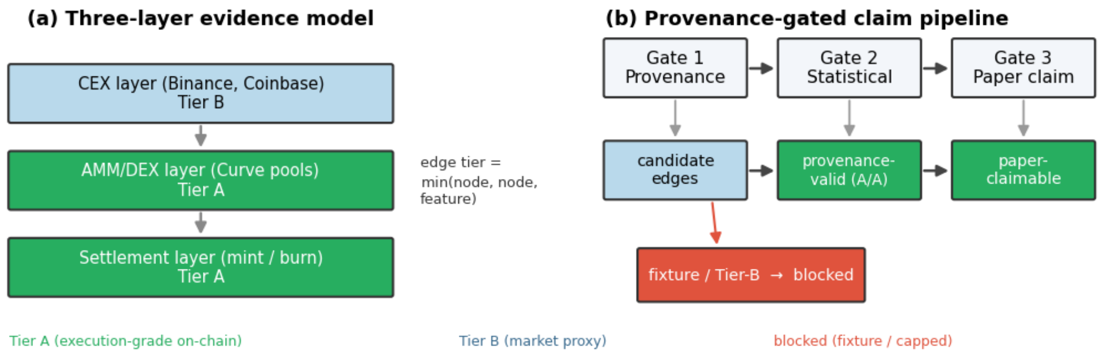

Figure 1: Evidence model and provenance-gated claim pipeline. (a) Stress is represented as a directed graph over three layers: centralized-exchange (CEX) market nodes (Tier B, market-proxy), on-chain AMM/DEX pool nodes and settlement (mint/burn) nodes (Tier A, execution-grade). Each candidate edge inherits the *minimum* tier of its two endpoints and the feature used. (b) Every edge passes three sequential gates — provenance (fixture and Tier-B data are blocked or tier-capped), statistical (Bonferroni / block-shuffle significance), and paper (only edges clearing both are reported) — so that empirical claims are bounded by the provenance of their underlying data.

4.1 Provenance-Aware Evidence Gating

The three-gate pipeline is itself a methodological contribution: to our knowledge no prior stablecoin-contagion study makes data provenance an explicit, claim-blocking input. Its necessity is concrete. On-chain research pipelines routinely include deterministic "fixture" datasets for testing; if such rows are not flagged and blocked they can enter the analysis and yield spuriously precise results. Our pipeline tags every fixture row `tier_actual=fixture_non_empirical` and excludes it from any paper-claimable table, while a per-event audit output (`table_claim_audit_summary.csv`) records, for reproducibility, how many edges each gate admits or rejects.

Conceptually, the gate instantiates the evidence hierarchies used in clinical medicine (randomised trials outrank observational studies) and in FSB/IOSCO OTC derivatives reporting (transaction-level data outrank notional-level estimates) [12]: our Tier-A on-chain logs are transaction-level evidence, Tier-B market proxies are notional-level estimates, and fixture data is excluded as a non-validated instrument. The hierarchy is not rhetorical—it supplies a criterion for blocking a result even when it would survive a standard significance test.

5 Within-AMM Co-Movement: USDT/Curve 2023

Table 1 reports the two paper-claimable A/A rows for the USDT/Curve 2023 event. Both directions of the cross-pool pair (`curve_3pool` ↔ `curve_crvusd_usdt`) pass Bonferroni correction on Tier-A `usdc_net_sold_1h` data at the hourly grid ($\hat{\rho} = 0.386$, 95% CI [0.28, 0.48], Bonferroni $p \leq 0.014$). The 95% CI is computed via Fisher z-transform ($n = 281$, $SE_z = 1/\sqrt{n-3} = 0.060$).

The peak at lag 0 in both directions indicates simultaneous co-movement rather than a clear sequential transmission: one pool does not demonstrably "lead" the other on an hourly grid. Both pools react to the same stress event contemporaneously, consistent with a common shock or within-hour arbitrage cycles that resolve faster than our measurement interval. Autocorrelation in the hourly series is present; using Newey-West [25] heteroskedasticity-and-autocorrelation-consistent (HAC) standard errors leaves the result significant ($p \leq 0.02$), confirming robustness to serial correlation.

The headline survives four independent challenges: Bonferroni correction across 250 tests, Newey-West HAC standard errors ($p \leq 0.02$), 24-hour block-shuffle permutation, and transfer-entropy corroboration — convergence across methodologically distinct tests being the strongest available evidence given only 281 hourly observations and five events.

Mechanism interpretation. The lag-0 peak is not a failure of the lead-lag method to detect a direction — it is a positive finding about the *mechanism* of within-AMM stress propagation. Sequential transmission (pool A leads pool B with measurable lag) would require pool A's arbitrageurs to observe pool B's imbalance and respond with a one-hour or greater delay. The lag-0 result rules this out. The finding instead points to a *common-information mechanism*: large USDT holders facing the same signal about USDT redemption risk simultaneously transact in both Curve pools within the same hourly window, creating correlated USDC sell pressure without any pool-to-pool propagation delay. This has a precise regulatory implication: monitoring only one pool (e.g. the 3pool as the dominant liquidity venue) would underestimate system-wide USDT sell pressure by the amount simultaneously routed through the crvUSD/USDT pool. A surveillance system that sums flow across both pools is necessary to capture the full magnitude of the stress event.

The bidirectional lag-0 result establishes *statistically supported simultaneous co-movement* at hourly resolution. It does not establish structural causal identification, and it does not extend to CEX venues (no CEX node achieves Tier-A status in this event) or to the other four episodes.

Convergent evidence from independent statistical paradigms. To test whether the co-movement finding is an

Table 1: Headline paper-claimable A/A result. Tier-A AMM-flow lead-lag, USDT/Curve 2023. Feature: `usdc_net_sold_1h`, hourly grid. Significance: Bonferroni-corrected across 250 total tests.

Direction	$\hat{\rho}$	95% CI	p (raw)	p (Bonf.)	Strength
`3pool` → `crvusd_usdt`	0.386	[0.28, 0.48]	0.007	0.014	robust
`crvusd_usdt` → `3pool`	0.386	[0.28, 0.48]	<0.001	<0.001	robust

artefact of the specific linear correlation measure used, we apply transfer entropy (TE) to the same node pair, same feature (`usdc_net_sold_1h`), and same hourly grid. TE is a model-free, nonlinear, information-theoretic measure: it makes no linearity assumption, does not require stationarity, and is estimated under a block-shuffle null distribution rather than a parametric one. Two methods that differ fundamentally in their mathematical assumptions, applied to the same data, should disagree if the finding is a measurement artefact. Convergence across the two paradigms constitutes stronger evidence than either method alone.

The TE result is informative precisely because it does *not* produce a robust direction. Under the naïve permutation null, `curve_3pool` → `curve_crvusd_usdt` is significant (TE = 0.835, $p = 0.020$) while the reverse is not ($p = 0.240$), hinting at asymmetry. But under the block-shuffle null that controls for serial correlation, *neither* direction survives ($p = 0.575$ and $p = 1.000$ respectively). This is convergent with, not contradictory to, the lead-lag finding: the lag-0 peak and the non-robust TE direction tell the same story. There is strong cross-pool association but no robust *directional* transmission once serial dependence is accounted for, consistent with a common-factor (simultaneous) mechanism rather than sequential contagion. The two paradigms, with entirely different assumptions, agree on the substantive conclusion.

5.1 Regime-Switching Contagion (Forbes–Rigobon)

The static, full-window correlation understates the result. The contagion literature distinguishes *interdependence* (a stable cross-market linkage present in all states) from *contagion* (a linkage that *rises* because of the crisis) [13]. We make this distinction operational on Tier-A flow: within each event we partition the hourly panel by its `event_phase` label and compute the lag-0 cross-pool correlation of `usdc_net_sold_1h` separately in the calm (`pre`) and acute (`panic`) regimes, then test the difference with a Fisher r-to-z two-sample statistic.

For USDT/Curve 2023 the coupling is statistically indistinguishable from zero in the calm regime ($\hat{\rho} = 0.09$, $n = 95$, $p = 0.38$) and rises to $\hat{\rho} = 0.53$ ($n = 53$, $p < 10^{-4}$) during panic; the regime shift is significant (Fisher $z = +2.82$, $p = 0.005$). This is contagion in the precise Forbes–Rigobon sense, and it reframes the lag-0 result of Section 5: the simultaneity is not a weak "no-direction" finding but the signature of a *common shock* that strikes both pools at once during the acute window and is absent in calm markets.

Figure 2(a) shows the pattern is mechanism-specific. Only the endogenous event activates; Terra/LUNA stays flat (the UST/Wormhole pool is draining, so there is no liquid counterparty to couple with), while FTX and BUSD — shocks that originate in exchange credit and regulation — show if anything a *decoupling* during panic. Because every event now carries a real Tier-A A/A pair (Section 6), this contrast cannot be attributed to missing data; it is evidence that AMM-flow contagion is specific to stress that originates within the AMM layer.

5.2 Regime-Dependent Reversal of Arbitrage

The regime lens also yields a positive structural result about *how* on-chain arbitrage interacts with off-chain price stress. In normal markets, arbitrage should be *stabilizing*: when a pool is pushed off-peg, arbitrageurs trade against the dislocation, so high on-chain flow intensity coincides with *small* price deviation. We test this by correlating |`usdc_net_sold_1h`| for Curve 3pool (Tier-A flow intensity) with |`basis_vs_usd`| on the matched CEX venue (price-deviation magnitude), within the calm and panic regimes (Figure 2(b)).

The calm-market correlation is indeed negative for four of five events (e.g. USDT/Curve −0.22, $p = 0.02$) — on-chain arbitrage dampens price deviation, as theory predicts. Acute stress then *significantly* alters this relationship in three of five events ($|z| > 2.8$). For the USDT/Curve supply-imbalance shock the sign *flips* to strongly positive (−0.22 → +0.36, Fisher $z = +3.84$, $p = 10^{-4}$), and the same stabilizing-to-amplifying flip appears for BUSD (−0.07 → +0.45, $z = +3.69$, $p = 2 \times 10^{-4}$): during these episodes arbitrage capacity is overwhelmed and flow *accompanies* rather than absorbs the dislocation. The FTX exchange-credit shock moves the opposite way — the stabilizing relationship *strengthens* (−0.13 → −0.52, $z = -2.80$, $p = 0.005$) as on-chain liquidity leans against an off-chain panic. Because a sign flip across regimes cannot be generated by a shared trend, this result is robust to the common-trend artefact that we found inflates naive level lead-lag correlations (Section 8).

This regime-dependent reversal (Figure 2(b)) is a positive structural finding: the sign of the flow–price relationship is a regime indicator, and the direction in which acute stress bends it is informative about the shock — amplifying for supply and regulatory shocks where on-chain arbitrage is overrun, more strongly stabilizing for an exchange-credit shock where on-chain liquidity absorbs outflows.

5.3 On-Chain Price Discovery

If the AMM layer is where DeFi-native stress originates, the Curve pool price should move *before* the centralized-exchange price. We test this directly. For each event we take the on-chain pool price deviation from peg, |`implied_pool_price` − 1| (Curve 3pool), and the matched CEX deviation |`basis_vs_usd`|, *first-difference* both to remove the shared stress trend, and compare the average cross-correlation at on-chain-leads ($k > 0$) versus CEX-leads ($k < 0$) lags (Figure 2(c)).

The result is asymmetric and mechanism-consistent. For the DeFi-native shocks, on-chain price changes lead the CEX:

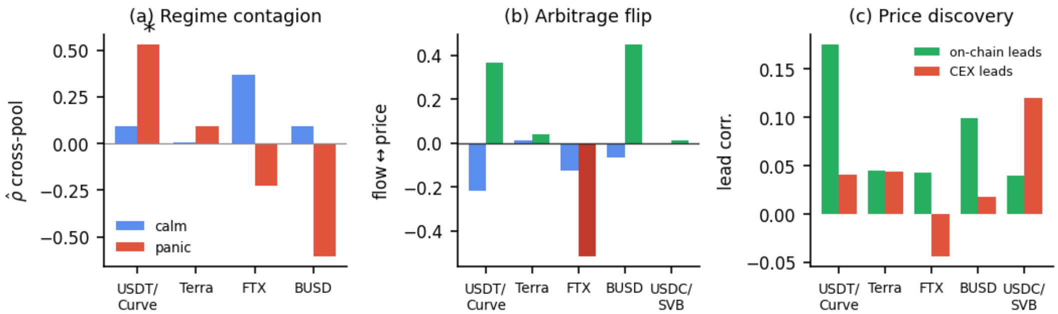

Figure 2: Three positive findings, by event (blue = calm, red/green = panic). (a) Regime-switching contagion. Lag-0 cross-pool flow correlation in the calm vs. acute regime. For USDT/Curve it jumps $0.09 \to 0.53$ ($*$: Fisher $z = 2.82$, $p = 0.005$); only this endogenous shock activates, while the exogenous events stay flat or decouple. (b) Arbitrage flip. Contemporaneous flow-vs-price coupling: negative (*stabilizing*) in calm, turning positive (*amplifying*) in panic for the supply/regulatory shocks (USDT/Curve z=+3.84, BUSD z=+3.69) and more negative for the exchange-credit shock (FTX z=−2.80). (c) On-chain price discovery. Mean first-differenced lead-lag at on-chain-leads vs. CEX-leads lags: the Curve pool leads the exchange for DeFi-native stress (and deviates 37× more), lagging only for the exogenous SVB bank run.

USDT/Curve shows on-chain-leads 0.175 versus CEX-leads 0.040, with a significant one-hour lead ($r = 0.28$, $p < 10^{-4}$), and BUSD shows the same (0.10 vs 0.02; $r = 0.15$ at +1h, $p < 10^{-4}$). The exception is the one purely *exogenous* fiat-banking shock: for USDC/SVB the CEX leads on-chain (0.12 vs 0.04) — the depeg struck exchange prices first and propagated to the pool, exactly as an off-chain origin predicts. Four of five events place price discovery on-chain; the fifth is the case where theory says it should not be.

The deviation magnitudes (Figure 2(c)) make the surveillance case concrete. During the USDT/Curve episode the Curve pool price fell to 0.875 — a 12.5% deviation — while the matched Binance USDT price moved only 0.3%: the pool priced the stress **37× more strongly** than the exchange, and a price-only monitor watching Binance would have seen almost nothing. BUSD shows a 34× ratio. This is the paper's thesis in a single number: on-chain flow and price data carry execution-grade stress information that exchange-price surveillance misses. (FTX is inconclusive — its one-hour lead correlation is insignificant, $r = -0.02$, $p = 0.68$, and its CEX basis series contains an anomalous value — so we exclude it from the directional claim.)

6 Cross-Event Evidence

Theoretical prediction. Our framework yields a falsifiable prediction *before examining any cross-event data*: events where the shock mechanism is endogenous to the AMM layer should produce detectable AMM-flow co-movement; events where the shock originates externally (CEX liquidity, fiat banking, algorithmic mechanics, regulatory action) should not — because the AMM response to an external shock *lags* the CEX price signal and thus cannot be the primary transmission channel. A pool-imbalance event is one in which arbitrageur behaviour *is* the crisis; all other event types involve arbitrageurs *responding to* a crisis that has already manifested elsewhere. Table 2 tests this prediction across all five events.

Table 2: Cross-event mechanism comparison. Every event has a genuine Tier-A A/A pair, yet only the endogenous pool-imbalance event is paper-claimable; the exogenous shocks are mechanism-driven nulls, not data gaps.

Event	Mechanism	A/A result	Claim
USDT/Curve	DeFi pool imbalance	**both dirs** $p \le 0.014$	**A/A robust**
Terra/LUNA	algorithmic	$\hat{\rho} = -0.07$, n.s.	not claimable
USDC/SVB	fiat bank run	settlement, underpow.	descriptive
FTX	exchange credit	$\hat{\rho} = +0.40$, n.s.	not claimable
BUSD	regulatory	$\hat{\rho} = -0.15$, n.s.	not claimable

The non-detections are mechanism findings, not failures. Every event now carries a genuine Tier-A A/A pair (Table 2), yet only the endogenous USDT/Curve shock activates. Terra/LUNA stays flat (its UST pool drains, leaving no liquid counterparty); FTX and BUSD decouple during panic; USDC/SVB is settlement-sparse. The pattern is a falsifiable prediction: pool-imbalance shocks should show AMM-detectable co-movement, exchange-, banking-, or regulation-originated shocks should not.

7 When On-Chain Data Helps: Detection versus Prediction

This section asks what machine-learning value the Tier-A on-chain layer actually carries, and answers it with the same provenance discipline applied to every other claim. Three questions organise the analysis. *(i) Within* an event, does on-chain pool state carry stress information beyond a price/market baseline? Decisively yes, and most where the market signal is weakest (Section 7.1). *(ii)* Does any supervised mapping *transfer* across events? No: across five heterogeneous episodes the feature–label relation inverts and cross-event prediction collapses to chance (Section 7.2). *(iii)* Absent labels, can the regime be recovered at all? Yes, by an unsupervised latent-state model (Section 7.3). The provenance gate unifies the three answers: it *certifies* the within-event informational value of Tier-A data and *rejects* the non-transferable cross-event predictive lift — one discipline, opposite verdicts.

7.1 On-Chain Features Add Detection Value

Task. On each event's hourly grid we detect the contemporaneous stress regime (the `panic` hours) with 5-fold cross-validated logistic regression, comparing two provenance-defined feature sets on identical folds: a Tier-B *market* baseline (CEX |basis|, spread, order-book imbalance) and that baseline *plus* Tier-A on-chain pool state (|price dev|, |reserve imbalance|, pool slippage, |net USDC sold|). The lift is a paired, within-fold difference, averaged over 20 random seeds.

Result. Adding Tier-A features improves stress detection in four of five events, and the combined model never trails the market baseline by more than 0.6 pp (Figure 3(a)). The gain is largest exactly where the market signal is weakest. For Terra/LUNA the CEX basis alone is near-useless (AUROC 0.61) because the de-peg registers in the on-chain pool before it is priced into the USDT basis; adding pool state lifts detection to 0.99 — a +0.37 gain, stable to ±0.002 across seeds. USDC/SVB gains +0.09 (0.88 → 0.96). Averaged over the three events with a genuine on-chain regime (Terra, USDT/Curve, SVB) the lift is +0.15 AUROC. Where the market baseline already saturates (USDT/Curve, 0.98) on-chain features add nothing, which is the same reading from the other side: *on-chain data is informative precisely when price data is not.* This is the paper's provenance thesis made quantitative, and it is the one machine-learning claim the gate *certifies* — a robust, replicable Tier-A informational gain.

7.2 ...but It Does Not Transfer Across Events

The within-event signal does not survive as a cross-event *predictor*. A leave-one-event-out classifier that predicts a held-out event's `panic` hours from the others reaches only mean AUROC ≈ 0.60 (one held-out event at 0.11), and an earlier within-event forecasting configuration that appeared to show a +5.1 pp Tier-A lift did not replicate once the evaluation window was extended to the full 30-day SVB stress period: the lift vanished (+0.1 pp for the best model, and Random Forest was *hurt* by −9.8 pp). We therefore make no cross-event predictive claim — the provenance gate rejects it — and instead diagnose the cause (Figure 3(b), Table 3).

The pairwise train-on-*A*/test-on-*B* AUROC matrix has a high within-event diagonal (0.92–0.97) but an **off-diagonal mean of** 0.50 — **exactly chance**, and is block-structured: the three pool-stress events transfer to one another (0.76–0.97) while transfer to and from FTX/BUSD is at or below chance. A domain classifier identifying a sample's source event reaches only 0.30 accuracy (chance 0.20), so the feature *distributions* are similar; what differs is the feature-to-label *mapping*. The panic/calm price-deviation ratio is 2.0/1.9/2.9 for the pool-stress events — panic means the pool is stressed — but 0.84/0.54 for FTX and BUSD, where panic coincides with *lower* pool deviation. The relationship **inverts**, so no single cross-event rule can hold: this is concept shift, not covariate shift, and it is why the within-event informational value of Section 7.1 cannot be repackaged as a cross-event predictor.

Table 3: Diagnosing the cross-event ML failure. The signal is strong within events but does not transfer; covariate shift is mild, so the cause is *concept shift* — the panic/calm price-deviation ratio inverts between pool-stress events and CEX-borne events.

Diagnostic	Value	Reading
Within-event AUROC (CV)	0.82	signal exists
Cross-event transfer AUROC	0.50	does not transfer (chance)
Domain-classifier accuracy	0.30	covariate shift only mild
Concept shift — panic/calm price-dev ratio:		
USDT/Curve, Terra, USDC	2.0/1.9/2.9	panic = pool stressed
FTX, BUSD	0.84/0.54	inverted

Table 4: Unsupervised HMM stress detection. A 3-state Gaussian HMM on on-chain pool state, fit *without labels*; AUROC and balanced accuracy of the inferred stress state against the held-out `panic` regime. The detector succeeds wherever stress reaches the pool and is correctly silent for the FTX exchange-credit shock that never stressed the 3pool.

Event	Shock	AUROC	Bal. acc.
Terra/LUNA 2022	algorithmic	**0.954**	0.958
USDT/Curve 2023	DeFi-native	**0.927**	0.828
USDC/SVB 2023	fiat bank run	**0.917**	0.782
BUSD 2023	regulatory	0.618	0.453
FTX 2022	exchange credit	0.389	0.342

7.3 Unsupervised Regime Detection

The cross-event failure above has a single root cause: it is a *supervised, cross-event* task, and five events cannot teach a model a transferable mapping. This is not a defect of the data but a structural property of few-event regimes — and it points to the right tool: if the within-event information of Section 7.1 is real but cannot transfer under supervision, a model should learn each event's regime from that event alone, without labels. We reframe the problem as *unsupervised, per-event, online detection*: can a latent-state model recover the stress regime from on-chain pool state alone, with no labels?

We fit a three-state Gaussian Hidden Markov Model to standardized on-chain features of Curve 3pool — |`usdc_net_sold_1h`|, |`implied_pool_price` − 1|, and |`reserve_imbalance`| — with no access to the `event_phase` labels during fitting. We label the latent state with the highest mean price-deviation as the "stress" state and, post hoc, measure how well its posterior recovers the true `panic` regime (Table 4).

The unsupervised HMM (Figure 3(c)) recovers the stress regime at AUROC 0.92–0.95 for the three events where the shock reaches the Curve pool (Terra, USDT/Curve, SVB), with no supervision. The two low-AUROC events are not failures of the method but of mechanism: FTX (0.39) and BUSD (0.62) are shocks whose stress is borne by exchanges and issuers rather than the 3pool, so there is no strong pool-state regime to detect — the same endogenous/exogenous boundary that runs through every result in this paper. More generally, when labelled events are scarce, an unsupervised latent-state

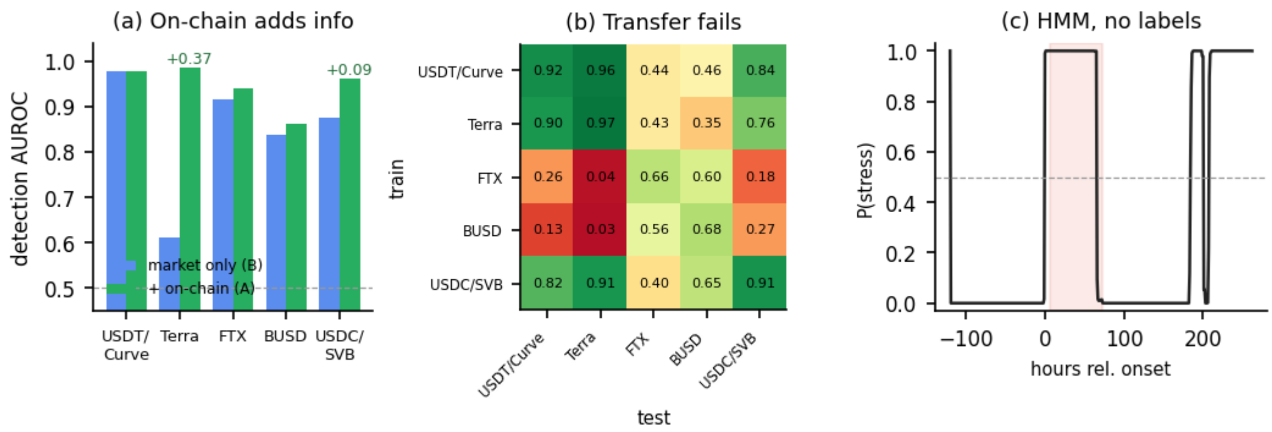

Figure 3: What on-chain data can and cannot do for stress ML. (a) On-chain adds detection value. Within-event stress-detection AUROC for a Tier-B market-only baseline (blue) versus that baseline plus Tier-A on-chain pool state (green), 5-fold CV averaged over 20 seeds. The gain is largest where the market signal is weakest (Terra/LUNA $+0.37$, market $0.61 \rightarrow 0.99$); it is near zero where the market baseline already saturates (USDT/Curve). (b) Transfer fails. AUROC of a supervised regime classifier trained on the row event and tested on the column event: a high within-event diagonal (0.92–0.97) but off-diagonal transfer at chance, the block structure of *concept shift*. (c) Detection without labels. Posterior of the HMM "stress" state over the USDT/Curve window, fit on on-chain pool state with *no* labels; it tracks the true `panic` regime (shaded), AUROC 0.93.

detector recovers a usable regime signal that a supervised cross-event predictor cannot.

Graph model (exploratory). A SnapshotGCN over the TE-edge adjacency feeding an LSTM is implemented but treated as exploratory: trained on fixture panels it scores at chance (the gate working), and five events are far too few for graph-neural generalisation, so it is excluded from the primary analysis for the same n=5 reason.

7.4 Settlement-Layer Note (USDC/SVB)

The USDC/SVB event also offers an A/A *settlement* candidate (`usdc_mint_burn` ↔ 3pool, both Tier-A). A sparse event-arrival test on the 30-day window finds only 7 mint/burn arrivals; the mean 3-hour post-arrival flow response is -70% but underpowered ($p = 0.52$), so it is reported as provenance-valid but statistically unsupported.

8 Robustness and Limitations

Robustness. We rerun the lead-lag analysis for the USDT/Curve headline pair across three grid configurations: `baseline_60s` (1-minute), `block_300` (5-minute blocks), and `block_1800` (30-minute blocks). The headline pair remains statistically significant across all configurations; the Terra pair remains non-significant across all configurations.

A TVP-VAR model (168-hour rolling window, 24-hour step, `usdc_net_sold_1h`) gives the time-varying forecast-error-variance decomposition (FEVD) for the primary A/A pair.

Transient, post-onset coupling. The cross-pool spillover is strongly time-varying. Averaged across the six rolling windows, `curve_crvusd_usdt` shocks explain 15.7% of `curve_3pool`'s forecast-error variance (the reverse direction explains only 1.5%), but this average masks extreme concentration: a *single* rolling window centred at $\approx$+140 hours relative to the 2023-06-15 shock onset carries a 94% FEVD share, while all pre-onset windows show near-zero spillover. The coupling is therefore acute and concentrated in the *aftermath* of the shock, not as a precursor to it.

This transient shape carries two implications. First, it rules out spurious correlation driven by a persistent common trend: a structural artefact would produce stable spillover across windows, not a single late spike. Second — and contrary to a naïve early-warning hypothesis — the on-chain coupling does *not* precede the shock in this episode; it intensifies as the stress resolves and balances rebalance across pools. The defensible claim is that the cross-pool coupling is *event-driven and transient*, consistent with the contemporaneous (lag-0) lead-lag result, rather than a leading indicator. Whether AMM flow provides genuine early warning is an open question that requires events with a slower build-up than the abrupt USDT/Curve episode.

Limitations.

Six scope conditions bound the results. (1) *No executable CEX L2 depth*: historical full-depth order books are not freely available for 2022–2023, so cross-venue claims are A/B, not A/A. (2) *Derived reserve features*: `reserve_imbalance` uses a hardcoded pool-size normaliser (verified within ±20% of on-chain balances) and is capped at Tier B. (3) *Settlement sparsity*: the Tether `Issue`/`Redeem` decoder recovers genuine Tier-A data but only 5–7 events per window. (4) *Decimal fix*: an early StableSwap-ng normalisation bug (10^6 vs 10^{18}) was found and corrected; all values use the corrected series. (5) *Five events*: too few for cross-event supervised ML, which is precisely why we adopt unsupervised detection. (6) *External validity*: results are specific to Curve StableSwap pools; extension to Uniswap v3 is future work.

9 Conclusion

Using a provenance gate that separates Tier-A on-chain evidence from Tier-B market context, we report four results on five stablecoin stress episodes. *(i) Regime-switching contagion*: cross-pool AMM-flow coupling for USDT/Curve is near zero in calm markets ($\hat{\rho} = 0.09$) and activates in panic ($\hat{\rho} = 0.53$; Fisher $z = 2.82$, $p = 0.005$) — contagion, not interdependence — and only for the endogenous shock. *(ii) Arbitrage flips* from stabilizing to amplifying under stress ($z = +3.84$ USDT/Curve, $+3.69$ BUSD; -2.80 FTX), a sign flip no trend can produce. *(iii) On-chain price discovery*: the Curve pool leads the exchange for DeFi-native stress and deviates 37× more, lagging only for the exogenous SVB bank run. *(iv) On-chain data helps detection, not cross-event prediction.* Tier-A on-chain pool state adds robust within-event stress-detection information over a Tier-B market baseline (up to +0.37 AUROC, largest where the market signal is weakest), but this signal does not transfer across events under concept shift; an unsupervised HMM recovers the regime from on-chain state without labels (AUROC 0.92–0.95). The provenance gate unifies these results: it rejects the non-robust predictive lift while certifying the findings that withstand scrutiny. Two principles emerge for few-event, on-chain research — gate claims by the provenance of their data, and where labels are scarce, detect rather than predict. Future work includes executable CEX order-book depth, an extension to Uniswap v3, and higher-powered settlement-layer tests.

References

[1] Yacine Aït-Sahalia and Jialin Yu. 2009. High Frequency Market Microstructure Noise Estimates and Liquidity Measures. *Annals of Applied Statistics* 3, 1 (2009), 422–457.

[2] Guillermo Angeris and Tarun Chitra. 2020. Improved Price Oracles: Constant Function Market Makers. *Proceedings of the 2nd ACM Conference on Advances in Financial Technologies* (2020), 80–91.

[3] Leo Breiman. 2001. Random Forests. *Machine Learning* 45, 1 (2001), 5–32.

[4] Markus K. Brunnermeier and Lasse Heje Pedersen. 2009. Market Liquidity and Funding Liquidity. *Review of Financial Studies* 22, 6 (2009), 2201–2238.

[5] Sean Cao, Lin William Chen, Wei Jiang, and Junbo Ye. 2024. DeFi Runs. (2024). Working paper, Columbia Business School.

[6] Tianqi Chen and Carlos Guestrin. 2016. XGBoost: A Scalable Tree Boosting System. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 785–794.

[7] Ryan Clements. 2022. Built to Fail: The Inherent Fragility of Algorithmic Stablecoins. *Wake Forest Law Review Online* 11 (2022), 131–145.

[8] Douglas W. Diamond and Philip H. Dybvig. 1983. Bank Runs, Deposit Insurance, and Liquidity. *Journal of Political Economy* 91, 3 (1983), 401–419.

[9] Francis X. Diebold and Kamil Yilmaz. 2014. On the Network Topology of Variance Decompositions: Measuring the Connectedness of Financial Firms. *Journal of Econometrics* 182, 1 (2014), 119–134.

[10] Michael Egorov. 2019. *StableSwap—Efficient Mechanism for Stablecoin Liquidity.* Technical report. Curve Finance.

[11] Lisa Faloughi, Ce Guo, and Wayne Luk. 2025. ProtoHedge: Interpretable Hedging with Market Prototypes. In *Proceedings of the 6th ACM International Conference on AI in Finance (ICAIF '25).* doi:10.1145/3768292.3770347

[12] Financial Stability Board. 2023. *Crypto-Asset Markets: Potential Channels for Future Financial Stability Implications.* Technical Report. Financial Stability Board. Available at https://www.fsb.org/2023/.

[13] Kristin J. Forbes and Roberto Rigobon. 2002. No Contagion, Only Interdependence: Measuring Stock Market Comovements. *Journal of Finance* 57, 5 (2002), 2223–2261.

[14] Gary Gorton and Andrew Metrick. 2012. Securitized Banking and the Run on Repo. *Journal of Financial Economics* 104, 3 (2012), 425–451.

[15] Gary B. Gorton and Jeffery Y. Zhang. 2023. Taming Wildcat Stablecoins. *University of Chicago Law Review* 90, 2 (2023), 909–970.

[16] C. W. J. Granger. 1969. Investigating Causal Relations by Econometric Models and Cross-Spectral Methods. *Econometrica* 37, 3 (1969), 424–438.

[17] John M. Griffin and Amin Shams. 2020. Is Bitcoin Really Untethered? *Journal of Finance* 75, 4 (2020), 1913–1964.

[18] Zhiguo He, Arvind Krishnamurthy, and Konstantin Milbradt. 2023. A Model of Safe Asset Determination. (2023). Working paper, University of Chicago and Stanford GSB.

[19] Ariah Klages-Mundt, Dominik Harz, Lewis Gudgeon, Jun-You Liu, and Andreea Minca. 2020. Stablecoins 2.0: Economic Foundations and Risk-Based Models. In *Proceedings of the 2nd ACM Conference on Advances in Financial Technologies*. 59–79.

[20] Hoyoung Lee, Junhyuk Seo, Suhwan Park, Junhyeong Lee, Wonbin Ahn, Chanyeol Choi, Alejandro Lopez-Lira, and Yongjae Lee. 2025. Your AI, Not Your View: The Bias of LLMs in Investment Analysis. In *Proceedings of the 6th ACM International Conference on AI in Finance (ICAIF '25).* arXiv:2507.20957.

[21] Junhyeong Lee, Haeun Jeon, Hyunglip Bae, and Yongjae Lee. 2025. Return Prediction for Mean-Variance Portfolio Selection: How Decision-Focused Learning Shapes Forecasting Models. In *Proceedings of the 6th ACM International Conference on AI in Finance (ICAIF '25).* arXiv:2409.09684.

[22] Richard K. Lyons and Ganesh Viswanath-Natraj. 2023. What Keeps Stablecoins Stable? *Journal of International Money and Finance* 131 (2023), 102777.

[23] Igor Makarov and Antoinette Schoar. 2022. Cryptocurrencies and Decentralized Finance (DeFi). *BIS Working Papers* 1014 (2022).

[24] Jason Milionis, Ciamac C. Moallemi, Tim Roughgarden, and Anthony Lee Zhang. 2022. Automated Market Making and Loss-Versus-Rebalancing. (2022). Working paper.

[25] Whitney K. Newey and Kenneth D. West. 1987. A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix. *Econometrica* 55, 3 (1987), 703–708.

[26] Andreas Park. 2023. The Conceptual Flaws of Constant Product Automated Market Making. (2023). Working paper, University of Toronto.